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SWIRL CUP MODELING – PART VI: DILUTION JET MODELING

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ABSTRACT

CFD simulations of gas turbine combustors, typically, do not include the inner and outer passages of the diffuser in the models for design analyses. This necessitates the specification of boundary conditions at the exit of the dilution holes into the combustor domain. This paper explores an approach for specifying this boundary condition that is applicable for combustor operating conditions at different power levels. First, the predictions for discharge coefficient are validated against the NACA benchmark data for different circular hole sizes and wall thickness. The agreement with data is within 5%. For this configuration, simulations are also performed to understand the effects of cross-flow on discharge coefficient and jet angle. Results indicate that these parameters are not very sensitive to the crossflow momentum ratio. Simulations are also performed for a CFM-56 swirl cup package in a 3-inch square flame tube. To understand the flow characteristics at dilution hole, the configuration included an outer flow passage. Results for this case indicate that distributions of flow variables at the dilution holes are two-dimensional and complicated due to existence of reverse flow regions. The swirling flow in the flame tube also has an impact on the dilution jet as the flow is subsonic. Since the flow variables distributions do not follow a pattern amenable for developing scaling rules, accurate modeling of dilution flow requires integrated diffuser-combustor analysis.

INTRODUCTION

Gas turbine combustion systems are being designed/optimized through the aid of CFD tools[1-3]. Though modern computers afford higher fidelity in CFD models in terms of resolving finer geometric and flow features, integrated modeling of diffuser-combustor system is still expensive for design analyses. Combustor modeling, then, requires specification of inlet conditions for swirlers, dilution and cooling hole boundaries. In rich-dome combustors, the penetration and mixing of primary and secondary dilution flows are critical to the predictions of emissions and pattern factors. The flow around the dilution jet is quite complex as it interacts with the dome swirling jet and the central recirculation bubble. The heat release often reaches a maximum just upstream of the primary dilution jets, resulting in high density gradients around the jets. Several studies on mixing of dilution jets have been reported[4-6] due to its importance for RQL concept to achieve low NOx emissions in rich-dome combustors. The equivalence ratio of primary reaction zone including the primary dilution air is adjusted to achieve complete combustion of fuel/air mixture. The dilution flow rates are also controlled to achieve a desired exit temperature profile.

For a typical combustor mesh with less than half a million nodes, the dilution holes are meshed with roughly 4X4 cells. The current practice is to assume a plug flow boundary condition over the effective diameter of the hole and the jet angle interpolated from data[10]. For this, the discharge coefficient is needed apriori and it is interpolated from experimental data. Since the dilution holes are on a curved surface and the dilution jets enter the combustor domain at an angle, calculation of boundary conditions becomes complex even for the plug flow assumption. For specifying a profile condition, at least a 10X10 mesh would be needed. This could be achieved by refining the coarse mesh locally in the dilution hole region. As will be shown later, the flow through the dilution hole is not symmetric about the center of the hole. Hence the boundary condition will not be a radial profile, but a two-dimensional mapping on a curved surface. This paper explores a methodology for specifying the dilution hole inlet conditions that is better than the current plug flow assumption. This problem has not been addressed in the literature.

The critical parameters in the flow through dilution hole problem are: pressure loss parameter (K) through the hole, hole size, wall thickness, boundary layer thickness, and cross-flow momentum ratio. The pressure loss parameter is defined as

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$$K = \frac{P_d - p_j}{P_d - p_d}$$

where subscripts d and j refer to the duct and jet conditions and P represents the total pressure and p represents the static pressure. CFD simulations are performed for flow through circular holes corresponding to the experimental conditions of Dittrich and Graves[7]. CFD predictions for discharge coefficient and jet angle have been validated for several K values and geometric configurations. The predictions are also compared with existing correlations[8]. A database of predicted flow conditions at the exit of the hole have been compiled. The flow conditions at the exit of the dilution hole are analyzed for identifying procedures to enable accurate modeling of dilution holes.

APPROACH

Fluent CFD code[9] is used to perform all the simulations presented in this paper. A sequential pressure-based algorithm is selected to solve the Favre averaged Navier-Stokes equations. Since the jet velocities would be high for high pressure ratio parameter cases, the compressibility effects are included in the model. Though Fluent offers several turbulence models, earlier studies have indicated that standard and realizable k-ε models provided the best results for the swirling and mixing flows. Compared to the standard k-ε model, the realizable k-ε model converges faster and hence it is selected for all the calculations.

VALIDATION

In order to validate the dilution hole effective areas, the Fluent predictions are validated against benchmark data of Dittrich and Graves[7]. In this NACA experimental study, the discharge coefficients for circular holes were measured for various flow parameters. The parameters considered were hole diameter, wall thickness at the hole, boundary layer thickness, and pressure ratio across the hole. The measurements showed that the corrected discharge coefficient correlated well with the pressure ratio parameter. The effects of boundary layer thickness and wall thickness were found to be small.

The computational domain for this case extended 12 diameters upstream of the hole and 5 diameters downstream of the hole. The height of the duct is taken to be 5 diameters to minimize any effect of the boundary layer on the duct wall above the hole. The plenum below the hole is made very large in order to prevent any influence of boundary condition on the flow near the hole. A total pressure boundary condition is applied at duct inlet. The total pressure can be calculated for a given value of pressure parameter (K). A static pressure condition is imposed at the free boundaries of the plenum below the hole as well as at the exit of the duct.

Grid Independence Study

First a grid sensitivity study has been carried out to select the optimum mesh size for parametric calculations. A hexahedral baseline mesh with about 600,000 cells was generated. The computational cells are clustered near the walls and inside the hole. Four cells were used to resolve the wall thickness of the hole. Two levels of adaptive mesh refinements were performed to generate additional meshes with velocity gradient as the adaptation criteria. In this study the value of K was taken as 10, the liner wall thickness and dilution hole diameter were taken as 0.04 and 0.75 inches, respectively. The computed mass flow rates through the dilution hole using the three levels of grids are shown in Table 1. The variation in mass flow rates is within 1%. For further study, depending upon the hole diameter, a typical baseline mesh size of 600,000 to 750,000 cells was used.

Mesh Size	Mass flow rate (Kg/s)
629095	0.00786
945406	0.00792
1011540	0.00779

Table 1 Mass Flow Rate Through the Hole as Function of Mesh Size

Validation of Discharge Coefficient and Jet Angle Predictions

In order to validate the discharge coefficient predictions, four different cases with values of K: 10, 20, 50, and 100 have been taken for which experimental data from NACA[7] are available. The theoretical mass flow rate through the dilution hole is calculated as the product of ideal jet velocity, density, and geometric area. Assuming isentropic flow, the ideal jet velocity and density are calculated from compressible-flow relationships using the duct total pressure and temperature. Table 2 shows a comparison between predicted and experimental values of discharge coefficients, the maximum deviation from experimental data has been observed to be less then 6% (for K = 10). Jet deflection angle

is another important parameter in the present study. The jet angle is calculated from mass averaged velocity components at dilution hole. The NACA experiment did not include measurements of jet angle. Kaddah[8] proposed correlations for discharge coefficient and jet angle as a function of K as[8]:

$$C_D = \frac{1.25(K-1)}{\left\{4K^2 - K(2-\alpha)^2\right\}^{0.5}}$$

and

$$\sin^2\theta = \frac{C_D}{C_{D\infty}}$$

where α is the ratio of mass flow rate through the hole to mass flow rate through the duct, and subscript α represents value when K tends to infinity. The jet angle is defined with respect to the plane containing the dilution hole. The jet angle predictions are compared with estimates from this correlation and the results are presented in tables 2 and 3. The CFD calculations underpredict the jet angle when compared with the estimates from the correlation.

K	C_D Fluent	C_D NACA	% Error	C_D Correlation
10	0.56	0.598	5.95	0.58
20	0.59	0.599	1.50	0.60
50	0.60	0.603	0.05	0.62
100	0.61	0.61	0.36	0.62

Table 2 Comparison of Discharge Coefficient Predictions with NACA Data

K	Jet Angle	Correlation
10	64.92	72.29
20	69.95	77.33
50	77.65	80.43
100	80.36	83.60

Table 3 Comparison of Jet Angle Predictions with Correlation

Effect of Crossflow

In order to study the effect of cross flow on jet exit angle and discharge coefficient, for a constant pressure ratio parameter (K = 20), three different values of cross flow momentum ratios (defined as ratio of jet momentum to crossflow momentum) are taken i.e. 0.1, 0.2 and 0.3. For these cases, dilution hole diameter is taken as 0.125 inches. The wall thickness has also been varied between 0.04 inch and 0.1 inch. In table 4 the first row corresponds to a case with no cross flow, and it has been included for the sake of comparison. It is noted that the discharge coefficient is almost independent of cross flow. The jet angle, however, decreases for low momentum ratios and remains constant for higher momentum ratios. The crossflow sets up a streamwise pressure gradient as it flows around the dilution jet. This causes additional deflection of the jet. It is clear that the effect of cross flow on jet angle is less prominent for the case of 0.1 inch wall thickness.

J	Wall thickness 0.04in		Wall thickness 0.1in	
	C_D Fluent	Jet Angle	C_D Fluent	Jet Angle
0	0.625	82.160	0.666	88.860
0.1	0.624	81.850	0.666	80.889
0.2	0.625	81.059	0.667	80.738
0.3	0.628	79.862	0.666	80.586

Table 4 Effect of Cross Flow on Discharge Coefficient and Jet Angle

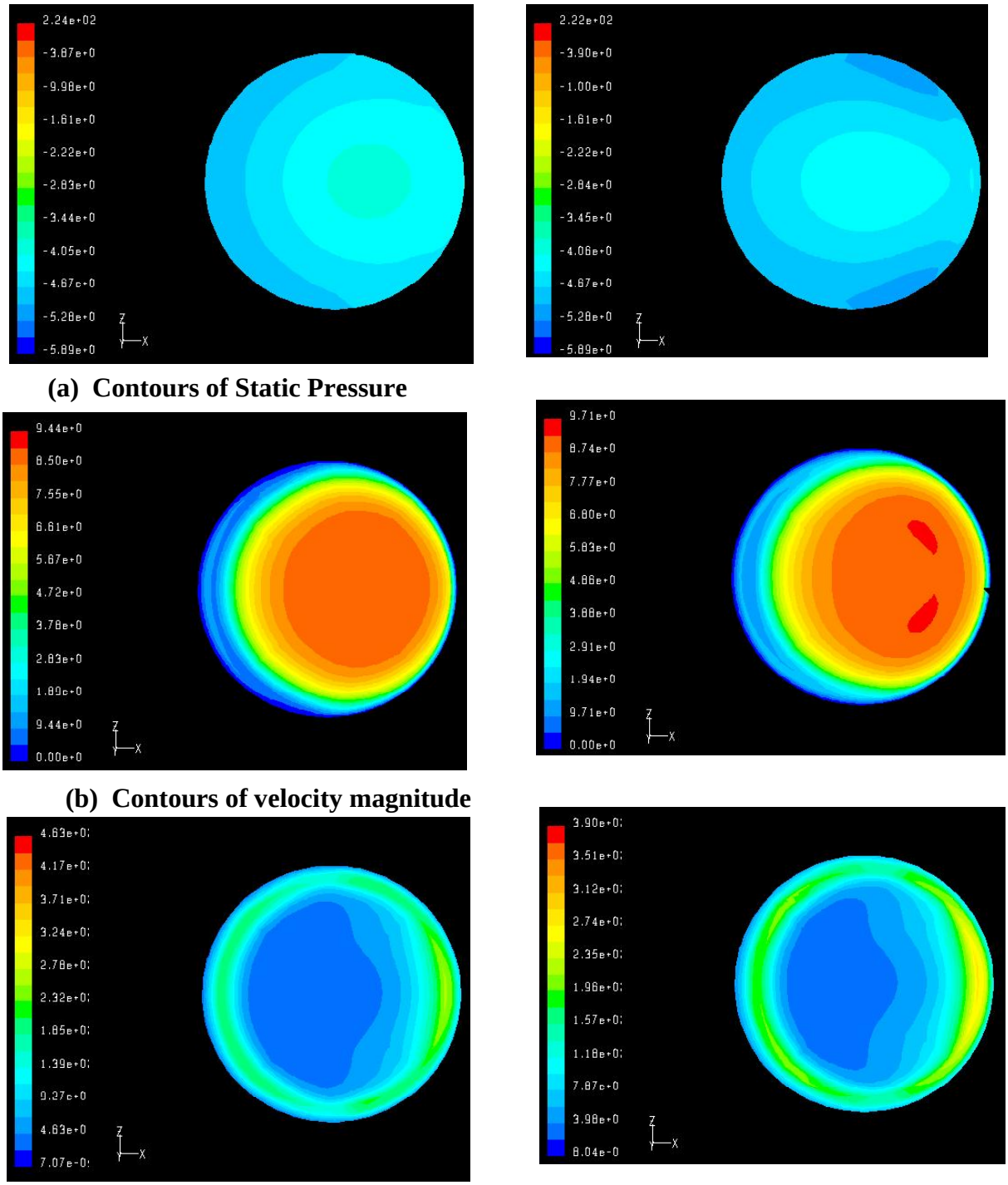
Effect of wall thickness and hole diameter

Present work has been extended to study the effect of liner wall thickness on discharge coefficient and jet angle. Two different values of wall thickness were taken i.e. 0.04 and 0.1 inch. For each wall thickness the hole diameter has been varied: 0.125, 0.25, 0.75 and 1.5inch. During this study the value of K was kept constant at 20 and the results are summarized in Table 5. It can be seen from this table that as wall thickness increases, discharge coefficient as well as jet angle increases. This is because increasing the wall thickness enables more pressure recovery after the flow makes the turn. It can also be noted that the increment in the discharge coefficient as well as jet angle decreases with respect to the hole diameter. Therefore, there should be an optimum wall thickness to diameter ratio for which maximum discharge coefficient and jet angle can be obtained.

Hole dia - thickness (in)	Jet Angle	C_D Fluent	C_D Correlation
0.125 - 0.04	82.160	0.625	0.601
0.125 - 0.1	88.860	0.666	0.601
0.25 - 0.04	77.880	0.662	0.601
0.25 - 0.1	84.100	0.692	0.601
0.75 - 0.1	74.680	0.647	0.601
1.5 - .04	65.230	0.564	0.601
1.5 - 0.1	68.820	0.583	0.601

Table 5 Effect of Wall Thickness and Hole Diameter on Discharge Coefficient and Jet Angle

J=0.0 J=0.3



(a) Contours of Static Pressure

(b) Contours of velocity magnitude

(c) Contours of turbulent kinetic energy

Figure 1 Flow Distributions at the Exit of the Dilution Hole for Cases with and without Crossflow

Flow Distribution at Dilution Hole Exit

In order to explore the possibility of scaling the boundary conditions, the flow through the dilution hole has been analyzed. The distributions of pressure, velocity magnitude and turbulent kinetic energy are shown in figure 1 for the cases of J = 0.0 and J = 0.3. Figure 2 shows the velocity distribution on a cross-sectional plane. As can be seen, flow is drawn from downstream of the dilution hole causing a reverse flow in the duct. This reverse flow becomes stronger as the pressure drop through the hole increases. Because of the high shear caused by this reverse flow, the turbulent kinetic energy is high in that region. Since the flow is unable to make a sharp 90-degree turn, there is also a reverse flow inside the hole for high pressure drops. The velocity magnitude peak on the exit plane occurs downstream of the geometric center of the hole. As shown in figure 3, the offset between the location of peak

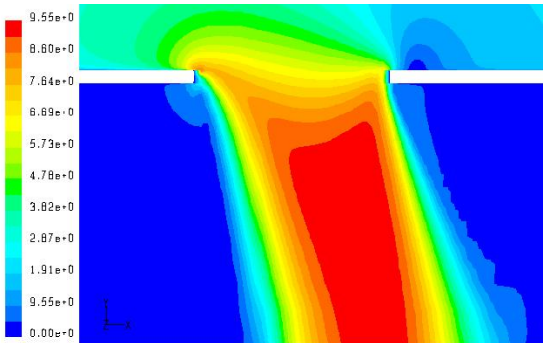


Figure 2 Velocity Distribution on a Cutting Plane Through Center of Dilution Hole for the Case of: Hole Diameter = 1.5”, Wall Thickness = 0.1” and No Crossflow

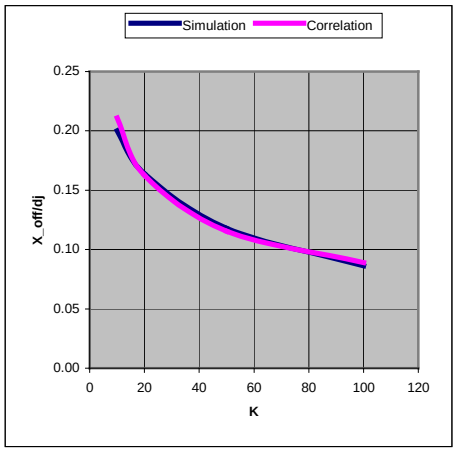


Figure 3 Variation of Dilution Hole Maximum Velocity Off-Center with Pressure Parameter

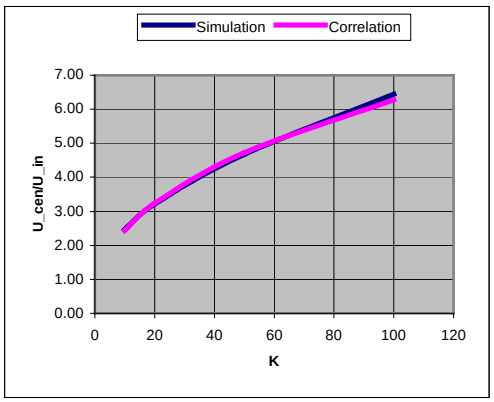


Figure 4 Variation of Dilution Hole Maximum Velocity with Pressure Parameter

velocity and geometric center of the hole decreases with pressure parameter. As expected, the magnitude of the peak velocity increases almost linearly with the K value.

Based on figures 3 and 4, several attempts were made to develop an equation for flow distribution at dilution hole exit. The independent variables in this procedure are the distance from the location of peak velocity and the angle from the duct streamwise direction. Though some success was realized for the velocity and pressure distributions, this procedure completely failed for describing the turbulence quantities. As can be seen in figure 5, the flow distribution is even more complicated for noncircular dilution holes used in the CFM56 combustor. It is concluded that the only way to adequately capture the details of the flow through dilution holes is to perform an integrated diffuser-combustor analysis.

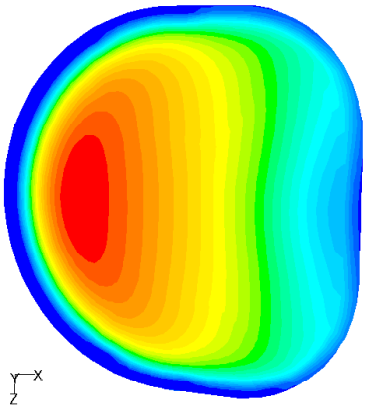


Figure 5 Velocity distribution at the Noncircular Dilution Hole of CFM56 Combustor.

DILUTION JET MIXING

Holdeman presented a review[5] of dilution jet mixing studies under subsonic cold and hot crossflow conditions. These studies have indicated that jet penetration and mixing were mainly influenced by momentum flux ratio, hole size, and spacing. To understand the mixing characteristics of the CFM-56 combustor dilution jets, a nonreacting flow simulation was performed for a simpler configuration of a CFM-56 swirl cup in a 3-inch square flame tube. This configuration corresponds to an experimental setup in which cold and reacting flow measurements were obtained[11,12]. Outer diffuser passage is modeled as a square duct around the flame tube. The computational domain for this case is shown in figure 6. A tetrahedral mesh with about 400,000 cells was generated for this model. The operating conditions for this study correspond to the maximum power condition of the CFM-56 combustor. The primary jets and secondary swirler inlet conditions were obtained from through-the-vane (TTV) calculations[13]. A uniform velocity condition was specified at the duct inlet. A fixed pressure condition was imposed at both the flame tube and duct outlets. The duct outlet static pressure was adjusted to get the correct amount of air flow through the dilution hole.

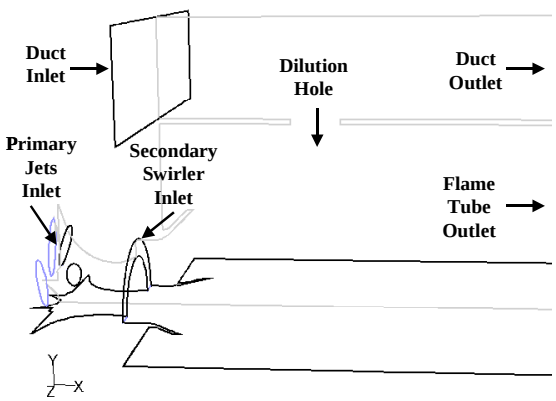


Figure 6 Geometry for the Dilution Jet Mixing Study

Figure 7 shows the axial velocity contours on a cutting plane through the center of the cup. As can be seen, there is a large central recirculation zone confined between the dilution and secondary swirler jets. Figure 8 shows the radial velocity distribution at the dilution hole exit plane. This velocity distribution is asymmetric about the duct streamwise direction and it is quite different from those shown in figure 1. This could only be due to the effect of swirling flow in the flame tube. Based on the calculated mass flow rate through the dilution hole, the discharge coefficient is predicted to be 0.63. For the calculated K value of 10 for this case, the predicted discharge coefficient is in good agreement with the data shown in Lefebvre's book[8]. Based on the velocity components at a location of maximum radial velocity in the jet, the jet angle (θ) is predicted to be 74°. Again this is in good agreement with the data shown in Lefebvre's book[8]. The jet penetration appears to be at least 3 diameters. Due to the presence of the recirculation zone, it is difficult to estimate the momentum flux ratio (J) relevant for jet penetration correlation. Crossflow velocity based on the total air flow from primary jets and secondary swirler and the cross-sectional area of flame tube is very low resulting in very high momentum flux ratio. If the crossflow velocity is taken to be the velocity just upstream of the jet, the momentum flux ratio is calculated to be 4. The correlation for jet penetration[8]:

$$\frac{Y_j}{d_j} = 1.15\sqrt{J}\sin\theta$$

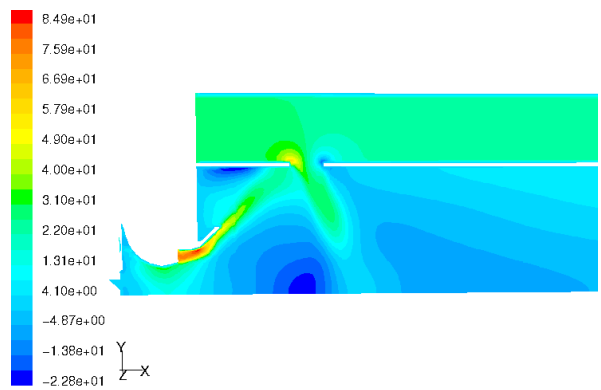


Figure 7 Axial Velocity Distribution on a Cutting Plane Through the Dilution Hole

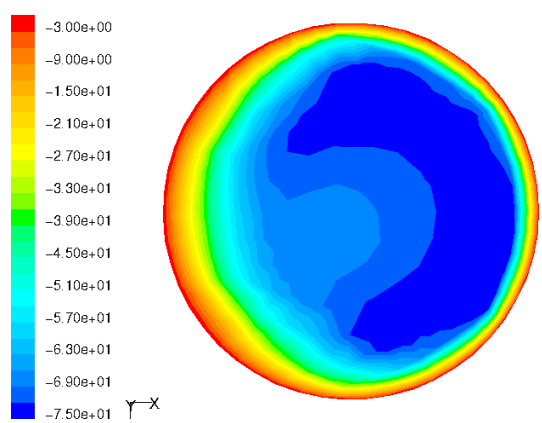


Figure 8 Radial Velocity Distribution on the Dilution Hole Plane.

yields a value of 2.2. Thus the jet penetration in this case is higher than the correlation value, but there is uncertainty in determining the appropriate J value.

To understand the extent of jet mixing, a scalar transport equation was solved. An additional equation for scalar fluctuations due to turbulence was also solved. The value of the scalar was set to unity at the duct inlet and set to zero at all other inlets. Earlier studies[2,14] have indicated that k-ε turbulence models underpredict scalar mixing. Reacting flow simulations performed for different combustors showed that combustor exit temperature profile predictions were in agreement with data when Schmidt number for scalar equation was set to 0.3. Hence this value is used in the results shown here. Figure 9 shows the scalar distribution indicating the penetration and mixing of the dilution jet. The maximum value of scalar in the bubble is about 0.5. The dilution air is observed to reach up to the venturi exit plane upstream of the jet. The scalar variance shown in figure 10 indicates large fluctuations in scalar concentration on the upstream side of the jet. The air flow from the primary jets and secondary swirler is squeezed between the dilution jet and the central recirculation bubble, creating high shear and turbulence kinetic energy (see figure 11) around the jet. This enhances mixing of dilution air into the bubble. The turbulent kinetic energy is observed to be maximum at the eye of the recirculating bubble.

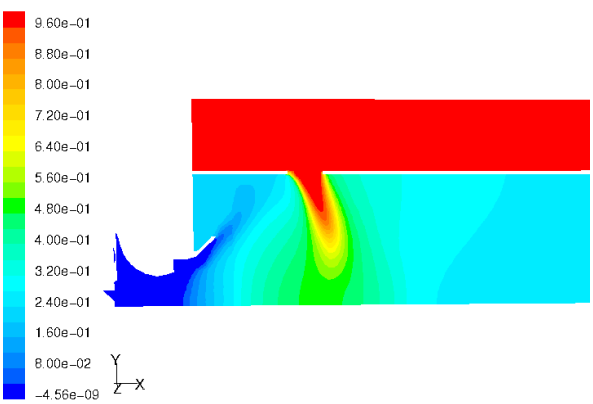


Figure 9 Scalar Distribution Showing Dilution Jet Penetration and Mixing.

Liscinsky et al.[6] define an unmixedness variable as:

$$\chi = \frac{\overline{C}_v}{\overline{C}(1-\overline{C})}$$

where $\overline{C}_v$ is the variance from the fully mixed state and $\overline{C}$ is the mean scalar value at the fully mixed

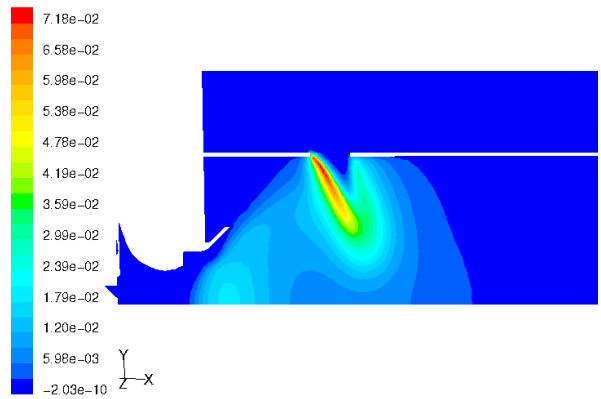


Figure 10 Scalar Variance Distribution Near the Dilution Jet.

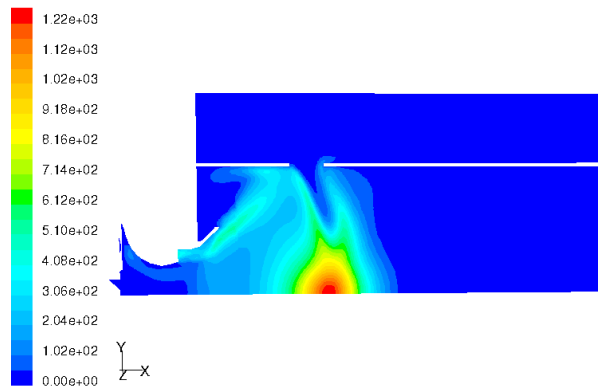
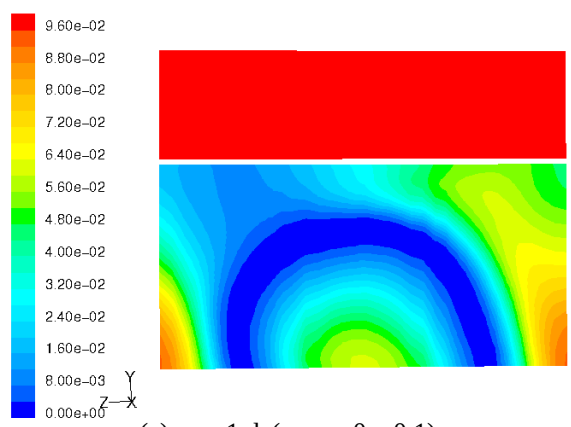


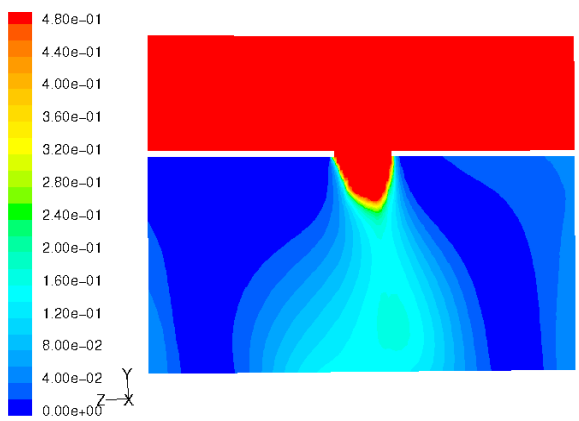
Figure 11 Turbulent Kinetic Energy Distribution on a Cutting Plane Through the Dilution Hole.

state. For the unmixedness variable, a value of zero indicates fully mixed state and a value of one indicates fully segregated state. Figures 12 (a) – (d) show the spatial distribution of unmixedness at several axial planes. To better visualize the mixing pattern, the ranges for different axial locations are not kept the same. It is clear from these figures that mixing is fast at the outer envelop of the recirculating

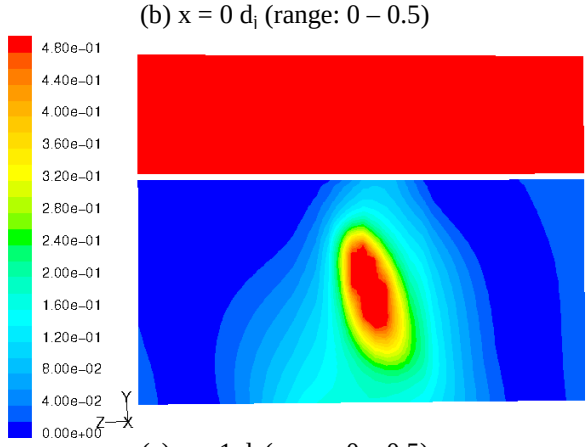


(a) x = -1 d_j (range: 0 – 0.1)

bubble. At an axial location of one diameter from dilution hole center, the unmixedness pattern has an elliptical shaped jet region and the variable reaches a maximum value away from the center of the flame tube. Eventually, the maximum value for unmixedness shifts to the center of the flame tube. Clearly, the mixing pattern is not symmetric about the center of the dilution hole due to the swirling flow. Results show that mixing is complete to within 1% only at five diameters from the dilution hole. Most of the GE rich-dome combustors have two rows of dilution holes, some have dilution holes on the first and third liner panels while others have them on first and second panels. For complete mixing, axial spacing between the rows of dilution holes needs to be five diameters for an inline pattern. For axial spacing smaller than five diameters, to improve the mixing pattern, the flow rates through the dilution holes could be adjusted to have lower jet penetration for the second row holes.

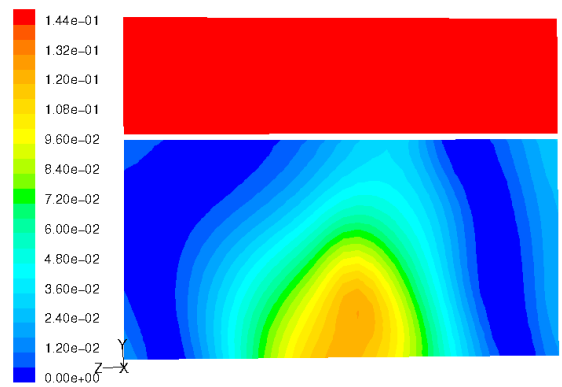


(b) x = 0 d_j (range: 0 – 0.5)

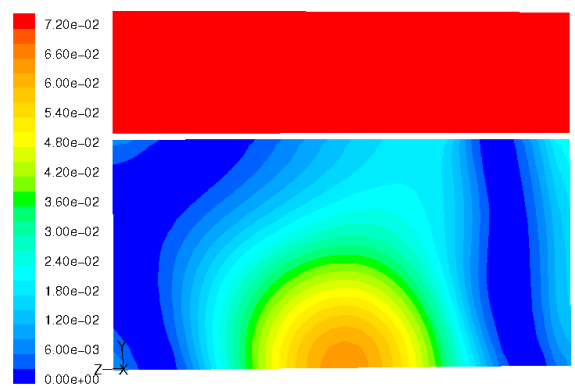


(c) x = 1 d_j (range: 0 – 0.5)

Calculations show that roughly 30% of the dilution air is mixed upstream of the jet. This provides an estimate for the amount of dilution air that participates in the combustion of the rich mixture from the swirl cup.



(d) x = 2 d_j (range: 0 – 0.15)



(d) x = 3 d_j (range: 0 – 0.075)

Figure 12 (a) – (d) Mixing Patterns at Various Axial Locations Relative to Dilution Hole Diameter.

CONCLUSIONS

In this paper, an attempt was made to develop a methodology for specifying the dilution hole inlet conditions that would be applicable for a wide range of combustor operating conditions. As a first step, the discharge coefficient predictions for various pressure parameters were validated with benchmark NACA data. The flow distribution for circular dilution holes was found to be complex due to existence of two recirculation regions in the proximity of the dilution hole. At high pressure ratio parameters, flow was drawn from downstream of hole in the duct creating high shear regions. For noncircular dilution holes, the flow pattern was even more complex. Since the jet Mach number is subsonic, the swirling flow within the combustor also has an impact on the dilution hole flow distribution. Attempts to develop a two-parameter mapping profile have failed. It is concluded that the only way to capture the details of the flow through the dilution holes is to perform integrated diffuser-combustor analyses. Such analyses have already been demonstrated[2,3,14]. The integrated analysis, however,

increases the mesh size of the model perhaps to more than a million cells. For reacting flow calculations, there are additional memory and processing overheads in the nonreacting regions of the diffuser with inner and outer passages. Local mesh refinement and parallel processing offers the hope of resolving all the details of flow through dilution holes and yet achieve an overnight simulation turnaround time so as to be relevant for design analyses.

The dilution jet mixing characteristics has been studied for the simpler configuration of a square flame tube. The curvature effect of annular combustors, hence, has not been modeled. Also, the effect of multiple holes has not been included in the present analysis. The results presented here indicate a significant impact of the crossflow that is swirling. In a reacting flow, the density gradients may also have an impact on the mixing patterns. Due to these reasons, the dilution jet mixing studies should be carried out for the annular combustor configurations with combustion model turned on. These studies are currently being pursued and the results will be published in the near future.

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